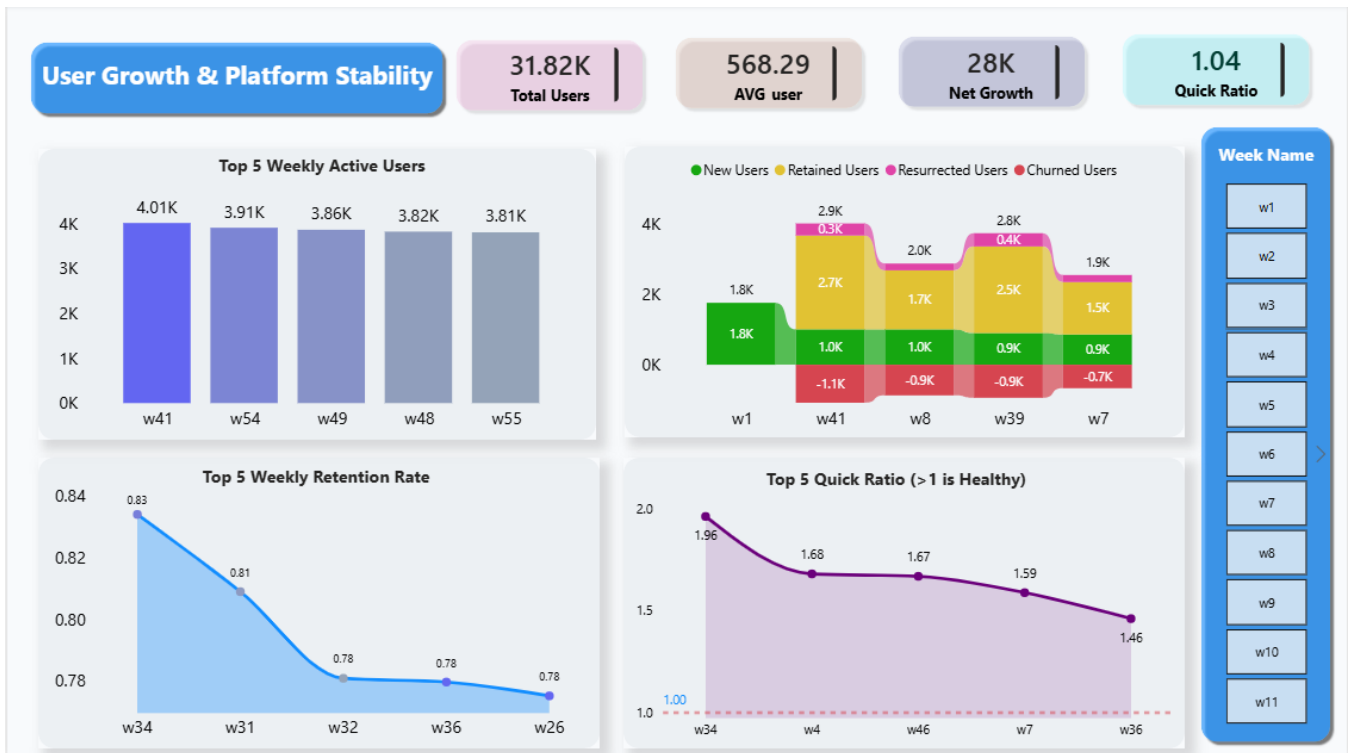


User Growth and Platform Stability



1. Core User Base & Acquisition

Total Historical Users: **31.82k**

- **What:** The unique count of every device that has interacted with the platform over 56 weeks.
- **Why:** This establishes the total "Market Reach" of the product.
- **How:** Cleaned and de-duplicated the device_id column to ensure non-breaking spaces and nulls didn't inflate the count.
- **Net Growth: 28k**
 - **What:** The total increase in the user base from the start of the observation period to the end.
 - **Insight:** This reflects a massive scaling phase, proving the product has high "top-of-funnel" attraction.

2. Engagement & Activity Benchmarks

- **Total Average WAU: 568.29**
 - **What:** The mean number of active users per week.
 - **Why:** While the total user base is 31k, this number tells us how many users are active in a *typical* week, helping with server capacity and resource planning.
- **Peak Engagement (W41): 4.01k Active Users**

- **Observation:** Week 41 stands as the record high for the platform.
- **Trough Engagement (W2): 1,654 Active Users**
 - **Observation:** The lowest point occurred very early in the project, indicating a successful recovery and growth trajectory since the beginning.

3. Growth Accounting & Sustainability

- **Average Quick Ratio: 1.04**
 - **What:** A metric comparing Acquisition (New + Resurrected) to Churn (Lost users).
 - **Why:** A ratio > 1.0 is the "Growth Threshold." At 1.04, the platform is officially growing sustainably, but it is in a "tight" zone where churn nearly matches acquisition.
 - **How:** Used complex DAX and SQL Window functions to segment users into New vs. Returning categories week-over-week.

4. User Loyalty & Permanent Churn

- **Unique Retained Users: 14,779**
 - **What:** Users who remained active for at least two consecutive weeks.
 - **Insight:** 46% of your total users show "loyal" behavior. This is the core audience that drives the platform's value.
- **Permanent Churn: 28,127**
 - **What:** Users who were active in the past but were **not** present in the final reporting week.
 - **Business Why:** This represents a "dormant audience." It is cheaper to re-acquire these users through "Win-back" campaigns than to find brand-new users.

5. Analytical Rigor (The Technical "How")

- **Data Sanitization:** Performed REGEXP filtering to remove "ghost" records and invisible whitespace that would have caused an ~8% error in growth reporting.

Volatility Analysis: Calculated a **Volatility Score of 4.31.**

- **Interpretation:** The platform has **Moderate Volatility**. Growth is not perfectly smooth (random/seasonal spikes exist), but it is consistently trending upward.
- **Interactive Visualization:** Developed a Power BI dashboard with a "Health KPI Card" that shifts colors based on the Quick Ratio to provide instant feedback to stakeholders.

Strategic Conclusion

The platform has successfully scaled to **31.82k** users with a healthy **Quick Ratio (1.04)**. However, with **28k users currently inactive**, the next phase of operations should focus less on expensive new acquisition and more on "**Resurrection Campaigns**" to bring the loyal core (14k+) back into the weekly active pool.

